Northwestern 9 at IslamicEval 2026: Fine-Grained Span Detection in Arabic Islamic Content Using CAMeLBERT-Mix

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Abstract

This paper describes our submission to the IslamicEval 2026 Task 1 (Span Detection) shared task, which aims to identify and classify evidence spans in AI-generated Arabic Islamic answers. We formulate the task as token-level sequence labeling and fine-tune CAMeLBERT-Mix using BIO tagging, overlapping-window inference, and prediction reconciliation to detect four evidence categories: Ayah, matn, isnad, and claimed_source. On the official development set, our approach achieves a Precision of 95.55%, Recall of 97.06%, F1-score of 96.30%, and Token Accuracy of 98.66%. Multi-seed and qualitative analyses demonstrate the robustness of the proposed framework for Arabic evidence span detection.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable performance in text generation. However, in sensitive domains such as Islamic question answering, they may generate hallucinated or unsupported statements, reducing the reliability of AI-generated responses.

Automatically identifying evidence spans is therefore essential for improving the transparency and trustworthiness of Arabic Islamic question answering systems.This work addresses IslamicEval 2026 Task 1 (Span Detection)(Alharbi et al., 2026), which aims to identify and classify evidence spans in AI-generated Arabic Islamic answers into 4 categories: *Ayah*, *matn*, *isnad*and *claimed_source*.We formulate the task as token-level sequence labeling using BIO tagging and fine-tune CAMeLBERT-Mix with overlapping-window inference and prediction reconciliation to improve span detection in long responses.

The primary challenge is accurate span boundary detection in long Arabic texts. On the official development set, our system achieves an F1-score of 96.30%, demonstrating the effectiveness of the proposed approach.

2 Related Work

Large Language Models (LLMs) have achieved remarkable success in text generation and question answering but may produce hallucinated or unsupported information, motivating research on hallucination detection, including recent shared tasks for Arabic Islamic content (Mubarak et al., 2025).

Pretrained Arabic transformer models such as AraBERT (Antoun et al., 2020), ARBERT, MARBERT (Abdul-Mageed et al., 2021), and CAMeLBERT have achieved strong performance across Arabic NLP tasks. Moreover, (Inoue et al., 2021) showed that pre-training data to fine-tuning data is is more important than corpus size, motivating our use of CAMeLBERT-Mix, which covers Modern Standard, dialectal, and Classical Arabic. Recent span-level methods focus on hallucination localization (Rykov et al., 2025), whereas our work addresses evidence span detection in Arabic Islamic content.

3 System Overview

3.1 Task Formulation

We formulate span detection as a token-level sequence labeling task. Given an Arabic question and its generated answer, the objective is to identify the exact boundaries of evidence spans in the generated response and assign each span to one of four semantic categories: *Ayah*, *matn*, *isnad*, or *claimed_source*.

CAMeLBERT-Mix was selected for its broad coverage of Modern Standard, dialectal, and Classical Arabic.

We employ the BIO tagging scheme, where the `B-` prefix marks the beginning of a span, the `I-` prefix marks subsequent tokens within the same span,

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and the O label represents tokens that do not belong to any annotated span. This produces nine token-level labels: O, B-Ayah, I-Ayah, B-matn, I-matn, B-isnad, I-isnad, B-claimed_source, and I-claimed_source.

3.2 Text Pre-processing and Label Alignment

The generated Arabic responses are tokenized using the fast tokenizer associated with CAMeLBERT-Mix. Character-level annotation offsets are aligned with the corresponding subword tokens using the tokenizer's offset mappings. Special tokens and padding tokens are assigned the value -100, ensuring that they are excluded from both the training loss and evaluation metrics.Responses longer than the model's maximum input length are divided into overlapping chunks. Each chunk contains at most 512 tokens, with a stride of 128 tokens between consecutive chunks. This overlap reduces the likelihood of losing evidence spans located near chunk boundaries. Padding is applied dynamically to the longest sequence in each batch.After inference, predictions from overlapping chunks are mapped back to their original character offsets and reconciled before span extraction. For tokens appearing in multiple overlapping windows, the corresponding prediction scores are aggregated to obtain a single final label. The resulting BIO sequence is then repaired to ensure valid label transitions, where invalid I- tags without a compatible preceding entity are converted into valid span starts, and inconsistent transitions initiate new spans. Finally, the repaired BIO labels are decoded into character-level evidence spans while removing duplicate predictions.

3.3 Overlapping-Window Prediction Reconciliation

Because tokens near chunk boundaries may appear in multiple overlapping windows, the same character range can receive more than one prediction. To reconcile these duplicate predictions, we group token outputs using the response identifier and the token's character-level start and end offsets. The probability vectors produced for identical token offsets across overlapping chunks are averaged, and the label with the highest averaged probability is selected as the final token prediction.The reconciled token predictions are then ordered by their character offsets and decoded into character-level evidence spans. Duplicate spans are removed, and leading or trailing whitespace is excluded from the predicted boundaries. This aggregation procedure prevents tokens appearing in multiple windows from being double-counted and provides a single consistent prediction for each character range.

3.4 Model Architecture

We fine-tune CAMeLBERT-Mix [1], a pretrained Arabic language model trained on a mixture of Modern Standard Arabic, dialectal Arabic, and classical Arabic data (Inoue et al., 2021).The contextual representation produced for each input token is passed to a token-classification layer that generates scores over the nine BIO labels. For an input sequence containing n tokens, the encoder produces contextual representations $\mathbf{H} = (\mathbf{h}_1, \ldots, \mathbf{h}_n)$. A linear classification layer maps each token representation $\mathbf{h}_i$ to a vector of label scores:

$$\mathbf{z}_i = \mathbf{W}\mathbf{h}_i + \mathbf{b}, \tag{1}$$

where $\mathbf{W}$ and $\mathbf{b}$ are trainable parameters. The predicted label for each token is obtained by selecting the label with the highest score. The model is fine-tuned jointly with the newly initialized token-classification layer.

4 Experimental Setup

To assess robustness, we trained the proposed framework using three random seeds (42, 52, and 62). All experiments were conducted on a Tesla T4 GPU (14.56 GB) using PyTorch 2.11 and Hugging Face Transformers. Each run required approximately 10.2 minutes and 3.44 GB of peak GPU memory.

4.1 Dataset Preparation

We use the official training and development data released for the IslamicEval 2026 Span Detection task. Each instance contains an Arabic generated response and character-level annotations identifying the evidence spans and their semantic types. The annotations are converted into BIO labels after tokenization.The original training and development partitions are preserved throughout the experiments. Long responses are represented by multiple overlapping chunks during training and evaluation. No additional external training data

[1] CAMeL-Lab/bert-base-arabic-camelbert-mix

Table 1: Training configuration.

Hyperparameter	Value
Pretrained Model	CAMeLBERT-Mix
Maximum Sequence Length	512
Sliding-window Stride	128
Training Batch Size	8
Evaluation Batch Size	8
Learning Rate	2×10^{-5}
Epochs	3
Weight Decay	0.01
Random Seed	42
Selection Criterion	Development F1

are used.Rather than introducing a novel architecture, we aim to establish a strong and reproducible baseline for IslamicEval 2026 through careful engineering choices tailored to Arabic span detection.The development set contains 484 responses with 2,728 annotated spans across four span types (Ayah, matn, isnad, and claimed_source). The average response length is 1,164.8 characters (median: 934), and 77.7% of the responses require chunking due to the model input length constraint.

4.2 Training Configuration

The system is implemented using PyTorch and the Hugging Face Transformers library. CAMeLBERT-Mix is fine-tuned for three epochs using a learning rate of 2×10^{-5}. The training and evaluation batch sizes are both set to 8, while weight decay is set to 0.01. We use a fixed random seed of 42 for reproducibility.

The development set was used for checkpoint selection, and final performance was evaluated on the test set. The corrected test run achieved an F1-score of 0.9126.

The model is evaluated after every training epoch, and one checkpoint is saved per epoch. The checkpoint achieving the highest development-set F1 score is restored at the end of training. Table 1 summarizes the main hyperparameters.

4.3 Evaluation Metrics

We evaluate the system using token-level precision, recall, F1 score, and accuracy. Tokens assigned the value `-100`, including special and padding tokens, are excluded from evaluation.

Precision, recall, and F1 are computed using micro-averaging over the eight entity-related BIO labels, excluding the `O` label. This prevents the large number of non-entity tokens from dominating the main evaluation scores. Token accuracy is calculated over all valid tokens, including those assigned the `O` label.

Table 2: Token-level performance on the development set.

Class	Precision	Recall	F1	Support
Ayah	0.9821	0.9889	0.9855	10905
matn	0.9679	0.9767	0.9723	11407
isnad	0.8794	0.9008	0.8900	2833
claimed_source	0.9365	0.9714	0.9536	5977
Overall	**0.9555**	**0.9706**	**0.9630**	–

Table 3: Development-set performance across three random seeds.

Metric	Mean	Std.
Precision	0.9583	0.0030
Recall	0.9721	0.0012
F1-score	0.9652	0.0019
Accuracy	0.9873	0.0006

The system was evaluated on the development set using token-level precision, recall, F1 score, and accuracy. Final performance on the official test set was evaluated using the shared-task F1 metric, with the corrected test run achieving an F1 score of 0.9126.

5 Results

Table 2 shows consistently strong performance across all four evidence categories. The model performs best on *Ayah* (F1 = 0.9855), while maintaining F1-scores above 0.89 for all classes, demonstrating strong generalization across different evidence types.

The original Task 1 submission received a score of 0.000 due to a submission error from our side. This result was used to rank our team in the official leaderboard and the shared-task overview paper. After resolving the submission error, the corrected run achieved an F1-score of 0.9126. This corrected result is reported to reflect the performance of our system and was not used for the official ranking.

Table 3 reports the average performance over three random seeds. The low standard deviations, particularly an F1-score of 0.9652 ± 0.0019, confirm the stability and reproducibility of the proposed framework.

5.1 Qualitative Error Analysis

To better understand the behavior of the proposed framework, we manually inspected the predictions on the development set. Table 4 summarizes the distribution of prediction outcomes.

As shown in Table 4, the model correctly

Table 4: Distribution of prediction outcomes on the development set.

Outcome	Count
Exact Match	2477
False Positive	242
Boundary Error	168
Missed Span	67
Type and Boundary Error	11
Type Error	5

Table 5: Representative error examples from the development set.

Error Type	Gold Type	Pred. Type	Span Example
Boundary Error	isnad	isnad	**Gold:** روى البخاري ومسلم عن ابن عباس رضي الله عنهما **Pred.:** ومسلم عن ابن عباس رضي الله عنهما
Type Error	Ayah	claimed_source	**Gold:** سورة آل عمران، الآية ١٣٠ **Pred.:** سورة آل عمران، الآية ١٣٠

predicts both span boundaries and evidence categories in most cases, achieving 2,477 exact matches. Most remaining errors are false positives (242) and boundary errors (168), indicating that span localization is more challenging than evidence type classification. In contrast, only five type errors and eleven combined type-and-boundary errors were observed, demonstrating reliable discrimination among the four evidence categories.

Table 5 presents two representative error cases. The first illustrates a boundary error, where the model predicts the correct evidence category but misses part of the annotated span. The second shows a type error, where the span boundaries are correctly identified but the evidence category is misclassified.

6 Conclusion

This paper presented our submission to the IslamicEval 2026 Task 1 (Span Detection) shared task. We formulated the task as token-level sequence labeling and fine-tuned CAMeLBERT-Mix using BIO tagging, overlapping-window inference, and prediction reconciliation. On the official development set, the proposed system achieved an F1-score of 96.30% and a token accuracy of 98.66%. The corrected test run achieved an F1-score of 0.9126, while the original submission score of 0.000 was retained for the official leaderboard ranking and shared-task overview paper. Qualitative analysis showed that evidence-type confusion is rare, while most remaining errors are related to span boundary localization. Furthermore, multi-seed evaluation confirmed the robustness and reproducibility of the proposed framework. Overall, the proposed approach provides an effective and reproducible baseline for Arabic evidence span detection. Future work will investigate CRF-based decoding, alternative overlap strategies, and improved BIO repair to further enhance span boundary localization.

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